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Peer Separation in Infant Chimpanzees, a Pilot Study

DEBORAH S. SNYDER,
University of Colorado School of Medicine
CHARLES E. GRAHAM, JAMES A. BOWEN
New Mexico State University
and MARTIN REITE
University of Colorado School of Medicine

ABSTRACT. Seven nursery reared chimpanzees (*Pan troglodytes*), 7.5 to 10.5 months of age, were studied to determine the effects of a short period of separation from their peers on behavior and heart rate. The chimpanzees were separated from their mothers at birth, and reared in the nursery in a group living environment. The experiment encompassed a 13-day period, including 4 days of normative baseline, 5 days during which three of the infants were separated and housed in isolation while the other four controls remained together, and 4 days in which all of the animals were reunited. Six quantified behavioral observations and five heart rate measurements were obtained daily. Following separation, the isolated infants were behaviorally agitated, and exhibited increases in vocalizations, rocking and self-clasp behaviors, as well as changes in facial expression including cry face, whimper face and pout face. Time spent locomoting decreased in all seven animals during the separation period. Agitated behavior in the separated and isolated infants alternated with stationary withdrawn behavior. Individual differences were prominent. Heart rate was notable by the generally poorly developed circadian rhythmicity throughout the 13 day period; significant HR changes did not appear otherwise associated with separation. Day three of separation appeared to represent a point of transition with stereotyped motor behaviors developing in the three isolated infants and in one control infant.

INTRODUCTION

Maternal separation experiments in macaque monkeys have proven to be a valuable animal model system for studying the effects of the disruption of attachment bonds, loss of mothering, and modeling certain aspects of loss-related affective disorders (REITE et al., 1981). The usefulness of such paradigms as animal model systems of relevance to man is in part a function of the ability to extrapolate from one species to another.

In comparison with monkey studies, relatively little has been published on experimental separations in great apes. Early studies suggested chimpanzees might be able to tolerate social isolation better than monkeys (DAVENPORT, MENZEL & ROGERS, 1966). However, GOODALL described profound behavioral disturbances, with sadness or depressive affective changes as major components, in orphaned wild chimpanzees, even though they were adopted and/or cared for by others (GOODALL, 1971a, b, 1979). The mortality in such orphaned chimpanzees has been distressingly high. More recently a peer separation experiment in 1.5- to 2-year-old chimpanzees was reported by BARD (1982), who found their "chimpanzees that were alone during separation reacted with high levels of 'protest' alternating with 'despair'

throughout the separation period." Similarly, recurring periods of agitated behavior have been described in human children during separation from the mother (MACCOBY & FELDMAN, 1972; ROBERTSON & BOWLBY, 1952, cited in BOWLBY, 1969). The present study was designed to evaluate the effects on behavior and heart rate of a short period of separation from peers in a group of nursery reared chimpanzee infants.

METHODS

The research was conducted at the Primate Research Institute, Holloman Air Force Base, New Mexico. Subjects were seven chimpanzee (*Pan troglodytes*) infants aged 7.5 to 10.5 months (see Table 1) that had been removed from their mothers at birth and raised in an infant nursery. None of the animals had experienced any other experimental manipulation prior to this study. The experiment encompassed a 13-day period, which included 4 days of normative baseline observations, 5 days during which three of the infants ('experimental') were separated and housed in isolation while the other four animals remained together in the nursery, and 4 days in which all of the animals were again reunited.

The routine daily nursery activity schedule is listed in Table 2. While lights were nominally out from 2100 to 0400, background illumination from adjacent rooms was sufficiently bright to permit necessary staff activity and viewing of animals. There was a great deal of physical contact between the caretakers and the infants, who not only were normally picked up for diaper changing, feeding, etc., but who were frequently picked up and held or cuddled as part of routine nursery caretaking. Apart from scheduled activities listed in Table 2, the infants were free to play and/or nap throughout the daytime hours. They were placed in sleeping cages in essentially random groups of two or three; they appeared to exhibit no strong preferences as to sleeping partners.

Table 1. Subjects used in this study.

Animal	Sex	Date of birth	Age at start of study	Separated or control
Perry	Male	Aug. 9, 1981	306	S
Skip	Male	Oct. 22, 1981	232	S
LeAnn	Female	Oct. 26, 1981	228	S
Kurt	Male	July 30, 1981	316	C
Desiree	Female	Aug. 31, 1981	284	C
Richie	Male	Aug. 11, 1981	304	C
Jacob	Male	Oct. 15, 1981	239	C

Table 2. Routine 24-hour activity schedule for nursery.

Time	Activity
0000	Sleeping (lights out).
0400	Lights on, morning awakening. Removed from sleeping cages, first morning feeding. Animals free to play/nap.
0730	Diaper change, daytime staff activity begins (cleaning etc.).
1000	Juice feeding.
1200	Diaper changing.
1630	Main meal of biscuits and formula.
2100	Diaper changing. Placed in sleeping cages in groups of 2 or 3, lights out.

Table 3. Specific behavioral variables discussed in this paper.*

Behavior	Definition
Locomotion and tandem walking	Walking, running, climbing or walking with an arm around another animal
A play	Playing alone
Social play	Play with other animals
Oral object exploration	Visual examination of inanimate objects usually while manipulating them
Oral self	Self-mouthing and self-licking
Rocking	Repetitive swaying body movements
Rest	At rest—no visible movement (eyes may be open or closed)
Contact others and enclosing and enclosed by	Combination of physical contact with or arms around another animal, and enclosed by another animal
Stereotypic motor activity	Atypical repetitive movements
Self clasp	Clasping one's arms around oneself
Piloerection	Hair standing straight up
Pout face, cry face, whimper face	As defined in BERDECIO & NASH (1981)
Hoot, cry, scream	Various vocalizations, ranging from a low "hoo" sound, to a high pitched scream

*All behaviors were expressed in terms of percentage of total time observed.

Quantified behavioral observations were obtained by a trained observer (D.S.S.) who watched each animal for a 6-min period, five times a day (at about 0800, 1000, 1200, 1400 and 1600 hours, observation order interchanged), dictating observed behaviors onto an audiotape. We utilized a behavioral taxonomy developed for macaque monkeys (see REITE & SHORT, 1980) which was modified to include behaviors appropriate for chimpanzees. The specific behaviors discussed in this paper are listed in Table 3. The audiotape was later played back and behaviors entered on a computer terminal; subsequent analysis provided duration in percent of total time observed (% TTO) and frequency (in counts per 1,000 sec) measurements of all observed behaviors (REITE & SHORT, 1983; WALKER & REITE, 1975). Additionally the observer, who was present throughout the day every day, took extensive handwritten descriptive notes.

Heart rate measurements were obtained five times daily at approximately 0700, 1100, 1500, 2100 and 0200 hours by the animal caretakers, who held the animal and counted the apical pulse with a stethoscope. Daily food intake was recorded. During separation, the three experimental infants were housed in separate $1.5 \times 1.5 \times 2.5$ m cubicles, which provided no visual contact with peers, caretakers or observers. The separated infants were able to hear human voices and other chimpanzees. The separated animals were observed through one-way glass. Each cubicle included a crib, blankets and toys. The caretakers entered the cubicles only to feed, change diapers and take heart rate readings. The four (control) infants remained in their normal social environment. Behavioral observations and heart rate measurements were continued on these infants as well. After five days of separation, all seven infants were reunited in their normal preseparation environment.

Behavioral data was analyzed by tabulating duration and frequency scores for each behavior for each day, and then grouping the days into baseline (four days), separation (five days) and reunion (four days), providing a single score on each variable, for each animal, for each experimental period. An analysis of variance with repeated measures was performed on mean

scores between conditions within experimental and control groups. All heart rate values were plotted for each animal, and a mean heart rate was obtained for each animal for each experimental period (baseline, separation and reunion).

RESULTS

Baseline Observations

During the four days baseline period the animals were all considered to be exhibiting their normal behavioral patterns by the attendants who had been in contact with them. There was considerable individual variability in terms of behavioral patterns and peer associations during the baseline time period.

Of the infants to be separated, *Perry* was the oldest. He was a very social animal, ranking highest in the dominance hierarchy of the entire group. *Skip* was the second oldest animal of the separation animals. He appeared to be a timid, quiet animal. He rocked frequently during baseline, primarily when a caretaker was in the room. He was neither very active, nor very social. The youngest animal to be separated was *LeAnn*. She preferred other chimps to the caretakers. She never ran to the caretakers when they entered the room, and did not appear to enjoy being held by the caretakers. Though a very social animal, she had no obvious pref-

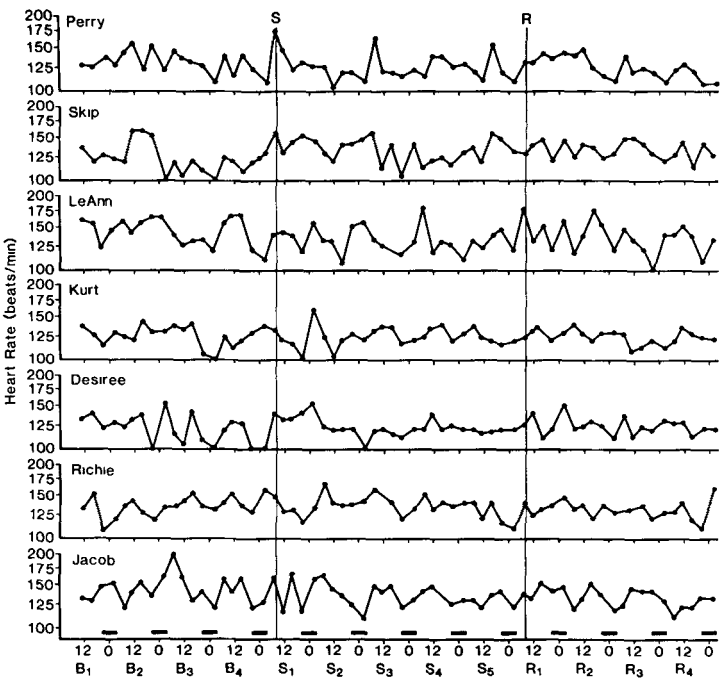


Fig. 1. Plots of all HR samples for all animals across the 13 day experimental period. Separation occurred at the first vertical dotted line marked S; reunion occurred at the second vertical line, marked R. Lights out periods are indicated by the dark markers on the X (time) axis. On the X axis B = baseline days, S = separation days, R = reunion days. The first three animals listed (*Perry*, *Skip* and *LeAnn*) were the experimental infants.

erences among the others. Rather, she appeared to be a dependent animal, and her constant contact with other animals appeared related to this.

Of the control animals, *Kurt* was the oldest (and the oldest of all seven). He spent a lot of time playing with others, and had the highest (30% of TTO) percentage of time spent rocking during baseline of any of the seven. *Richie*, quite social and the second oldest of the "control animals," was playful. *Desiree* spent more of her time with the caretakers than with other chimps and was considered as a "peoples chimp" and "spoiled" by the caretakers. *Jacob*, physically the smallest, received more play overtures and social contact from other animals than he initiated. He seemed to be very dependent on the caretakers, and would often begin rocking when the caretakers were in the room if they did not quickly pick him up. The animals who rocked frequently during baseline, generally appeared to increase their frequency of rocking in the presence of a caretaker if not picked up. Heart rate values obtained during baseline (as well as during the separation and reunion periods) are plotted in Figure 1. Noteworthy is the generally poorly developed circadian rhymicity. Evidence of the normally expected circadian rhymicity of HR is apparent in only about half of the baseline days.

SEPARATION

The three experimental infants were separated between 0745 and 0815 and placed in their isolation cubicles. Each infant was removed from the group separately, at 15-min intervals, so that behavioral observations could be collected immediately. *Skip* was separated first, followed by *LeAnn* and then *Perry*. Following separation, all experimental infants were behaviorally agitated, and exhibited increases in vocalization, which included hooting, crying and screaming. Rocking behavior was increased, as well as self-directed behaviors such as self-clasp. Individual variability was considerable during the period of separation in both the experimental and in the control animals, as will be discussed below. Examining group mean values for behavioral change, however, activity measures (locomotion and tandem walk) were significantly decreased ($F = 12.84$, $df = 2, 10$, $p < .05$) in both the experimental and the control animals (Fig. 2, top); time spent resting was increased (not significant) during the period of separation in both groups (Fig. 2, center). Self-play was significantly diminished in the separated animals ($F = 31.50$, $df = 2, 10$, $p < .05$), and increased slightly in the control animals (Fig. 2, bottom). Vocalizations increased during separation in all three separated infants; control animals did not vocalize (Fig. 3, top). Rocking, variably present in both groups during baseline, increased in two of the three separated infants during separation (Fig. 3, center). Self-clasp behavior increased in all three separated infants during separation, and in one control infant (Fig. 3, bottom). These differences were statistically significant ($F = 25.39$,

Table 4. Mean values (±S.D.) for all HR samples during baseline, separation and reunion.

	Baseline	Separation	Reunion
Perry	131±13	128±18	125±14
Skip	125±17	134±15	133±11
LeAnn	145±19	134±12	138±21
Kurt	125±12	125±12	120±21
Desiree	122±17	123±11	124±11
Richie	136±13	136±14	132±11
Jacob	143±20	136±15	132±12

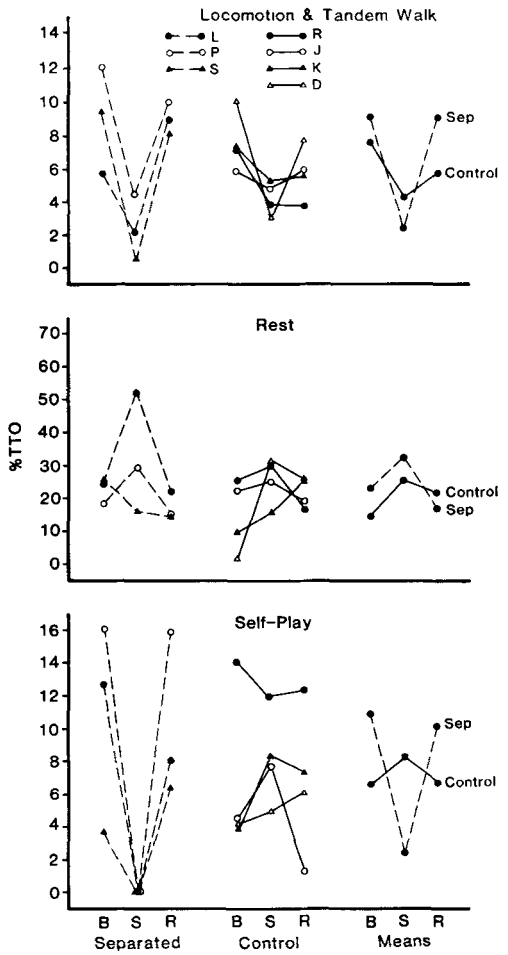


Fig. 2. Scores of the percentage of total time observed (% TTO) for three behaviors, Locomotion and Tandem Walk (top); Rest (center), and Self-Play (bottom) during baseline (B), separation (S) and reunion (R) for the three separated infants (left graphs), the four control infants (center graphs) and the mean scores for separated (sep) and control infants (right graphs). L = *LeAnn*; P = *Perry*; S = *Skip*; R = *Richie*; J = *Jacob*; K = *Kurt*; D = *Desiree*.

$df = 2, 10, p < .001$). Food intake generally remained consistent throughout the experiment for both groups of animals.

During separation, all three experimental animals exhibited facial expressions including cry face, whimper face and pout face (as described by BERDECIO & NASH, 1981) which they had not exhibited during baseline. These facial expressions were not observed in the control animals.

Mean HR values for the baseline, separation and reunion periods are listed in Table 4. No consistent changes were noted in HR during the separation period in either the experimental or the control animals. Two of the infants had a slightly lower mean HR during separation than during baseline. These were *LeAnn*, whose mean separation HR was 134, compared to a

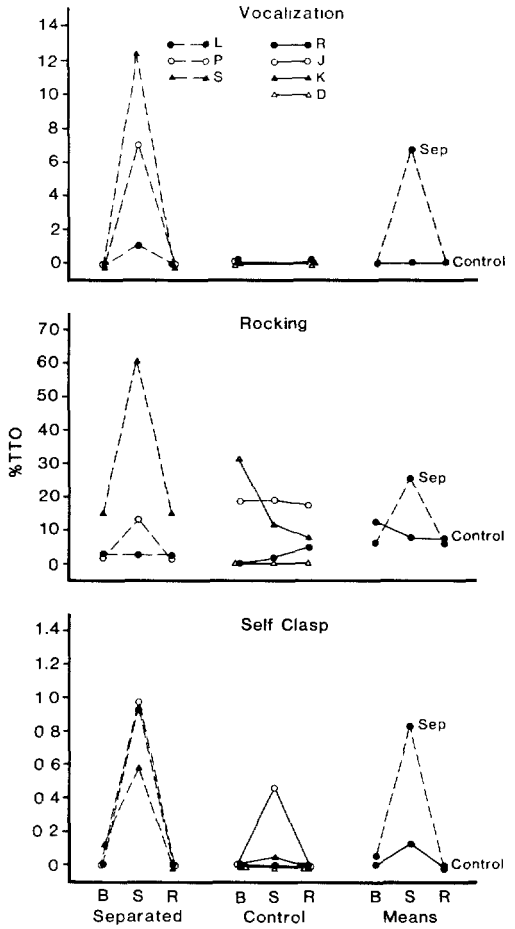


Fig. 3. Scores for vocalization (top), rocking (center) and self-clasp (bottom) behaviors, in % TTO, as in Figure 1.

baseline mean of 145, and *Jacob* (a control) whose baseline mean was 143, and whose separation period mean was 136. While the three experimental infants showed greater changes from baseline during separation and isolation than did the four control infants, these changes were not statistically significant.

Individual differences during separation were prominent. *Skip*, separated first, began screaming and rocking immediately. He exhibited a whimper face almost constantly during his rocking bouts. These episodes of rocking and screaming alternated with sleeping. By the third day of separation *Skip* was exhibiting episodes of violent behavior characterized by banging on his crib and repeatedly slamming the sliding door on his crib open and shut. During this time he also began self-clasping, and engaging in more oral-self behavior. The violent behavior subsided during the last two days of separation, and screaming bouts were limited to the period of time immediately following feeding and diaper changing.

Perry, separated second, initially reacted violently—screaming, locomoting and throwing himself against the glass walls of the isolation cubicles. He rocked when he wasn't locomot-

ing. The remainder of the first day of separation he was less active while awake, screaming after the caretakers left him and rocking intermittently with frequent napping. By the third day of separation, he was spending a great deal of time sitting in the corner of his crib quietly picking up a toy and dropping it, and he began exhibiting a pout face. Early in separation, the caretakers noted that *Perry* seemed to be shaking when they picked him up and he would not release them when they tried to put him down after feeding or changing. Later during separation, the caretakers noted that when they put him back in his crib following feeding or changing, he would let go and fall back limply.

LeAnn was asleep when removed from the social group and placed in isolation. Separated last, she began screaming when she awoke about 15 min later and found that she was alone. She appeared to be the least agitated of the three separated infants. She exhibited considerable rocking and oral object exploration of the crib the first day of separation, with intermittent napping. By the third day, she was spending most of her waking time sitting, staring into space, or lying quietly. At this point, she developed a stereotypy of holding her arm extended motionless for periods of up to several minutes at a time.

Among the control animals, *Kurt* exhibited no prominent behavioral changes during the separation period, seemingly equally comfortable and social with the remaining three animals. *Richie* appeared actually to be happier than he had during baseline, as with *Perry* gone, he was the most dominant animal of the remaining four. *Desiree* began to suck her thumb frequently on the third day of the separation. She had not been observed to do this during baseline. *Jacob* exhibited increased self-clasp behavior during the separation period; his behavior was not otherwise noticeably altered.

Among the three experimental animals, agitated behavior seemed to recur at intervals throughout separation, alternating with stationary, withdrawn behavior. Agitation often followed feeding, diaper changing, or being put down after heart rate readings.

Reunion

Activity levels were high the first day of reunion, as all animals spent the day running, playing and contacting each other. The two females (*Desiree*, a control and *LeAnn*, experimental) clung to each other more than they had during baseline. During the first day of reunion, *LeAnn* engaged in frequent temper tantrums when another chimpanzee broke contact with her. Social play and contact scores were generally higher during the first day of reunion than during baseline. Again, no noticeable HR changes accompanied reunion. Most behavior scores returned to baseline values following reunion.

DISCUSSION

These findings demonstrate that a short period of peer separation (and isolation) in infant chimpanzees may be accompanied by marked behavioral changes. We found a decrease in locomotion, and an increase in rest behavior. These changes of course could have been influenced by the alteration in the physical environment, with less opportunity for locomotion. Atypical or unusual behaviors, which included vocalizations, rocking and self-clasping, increased in the experimental group. The motor stereotypy developed by *LeAnn* on the third day of separation is similar descriptively to the "floating limb" phenomena described in iso-

lated rhesus monkeys by FOLEY (1934) and CROSS and HARLOW (1965). Our control group also experienced a separation from their then isolated friends, and one might, therefore, have expected to see some response, albeit lesser, in these animals. Two of the control infants (*Desiree* and *Jacob*) did in fact exhibit behavioral changes, including thumb sucking and self clasping. It is of interest that both thumb sucking and rocking have been described in human children as a response to separation which included placement in a new environment (HEINICKE & WESTHEIMER, 1965; PRUGH et al., 1953). Following reunion, there was evidence of increased contact between the experimental and control animals.

Significant changes in heart rate were not noted to accompany separation. Specifically, we found no evidence of significant bradycardia, which often accompanies maternal separation in macaque infants (REITE et al., 1981). *LeAnn* exhibited the greatest decrease in HR during separation, but also rested the most (see Fig. 3). It is possible that the HR decrease simply reflected the decrease in activity. *Skip*, who rested less during separation than during baseline (see Fig. 3), had an increase in mean HR during separation (Table 4). Such an influence would likely not have been operative in the case of *Jacob*, however. Bradycardia is not a universal accompaniment of separation. It may not be observed in human children who experience brief separations from their mothers during the birth of a sib (FIELD & REITE, in press). In young monkeys it is observed primarily in infants who remain in the social group when separated (and who become depressed); it is much less common in infants housed alone or in isolation when separated (REITE et al., unpub. observ.).

Notable was the generally poorly developed circadian HR rhythm in the animals throughout the 13 day period. This observation suggests that the nursery environment may not have provided sufficiently strong circadian synchronizers, or "Zeitgebers," to strongly entrain the usual 24-hr rhythm. It would be interesting to know whether the relatively weak circadian periodicity extended to other biological rhythms as well, such as body temperature and steroid hormones. An important caveat is that the method for obtaining HR samples was far from ideal, as any change, e.g., acceleration or deceleration, induced by the measuring procedure per se, i.e., picking up the infant, would contaminate the result. Telemetric methods would have been far superior, but were not available.

The recurring instances of agitated behavior in the separated chimpanzees were seen following feeding, diaper changing and heart rate readings. Since these duties were performed by the caretakers, it appears that following brief contact with (and renewed separation from) the caretakers, the animals became agitated. Similar reactions have been noted in human children. HEINICKE and WESTHEIMER (1966), in a study of ten children who were placed in a nursery, found that following visits by fathers, the children would react with screaming and crying similar to their initial reaction to separation. PRUGH et al. (1953) described episodic aggressive behavioral episodes as frequently accompanying separation and hospitalization in human children.

These were quite young, nursery reared infants. It is likely therefore that they had a major attachment bond to the caretakers who raised them in addition to whatever peer bonds they had, or were forming. The disruption of this bond, not complete since caretakers did visit to perform necessary chores, may have constituted a major stress.

Noteworthy is the fact that the 3rd day of separation appeared to represent a transitional point between initial agitation, where present, and the development of somewhat more atypical behaviors, including repetitive toy dropping and a pout face in *Perry*, episodes of violent

behavior in *Skip*, a "floating-limb" like phenomena in *LeAnn*, and thumb sucking in *Desiree*. These might all be classified as forms of motor stereotypy.

In this study, the separated animals were also isolated and in a strange environment. It has been shown that isolation, even for short periods, can result in atypical motor behavior, including stereotypies (DAVENPORT, MENZEL & ROGERS, 1966). We cannot say how much of the behavioral changes in the experimental infants were due to separation, and how much to isolation; for this reason it would appear unwise to extrapolate these limited pilot findings too far. What is clear, however, is that this experiment suggests that peer (and caretaker) separation and isolation in young chimpanzees does result in significant behavioral alterations, with marked individual variability. Evidence of alteration in physiological systems is not clear. A more definitive study, possibly involving a maternal separation paradigm, and during which multivariable physiological data could be obtained, would appear warranted.

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Authors' Names and Present Addresses: DEBORAH S. SNYDER, *College of Veterinary Medicine and Biomedical Sciences, Colorado State University, Fort Collins, Colorado 80523, U.S.A.*; CHARLES E. GRAHAM and JAMES A. BOWEN, *Primate Research Institute, New Mexico State University, Las Cruces, New Mexico 88330, U.S.A.*; MARTIN REITE, *Department of Psychiatry, Primate Laboratory, University of Colorado School of Medicine, Denver, Colorado 80262, U.S.A.*